

A Short Review of Matrix Algebra

The language of multivariate analysis

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Reading: Zelterman (2015), Ch. 4; Izenman (2008),
Secs. 3.1–3.2

Agenda

Data and Summaries

Transformations and Direction

Volume and Dimensionality

Quadratic Forms and Distance

Partitioning and Synthesis

Why Spend Time on Matrix Algebra?

Suppose each customer has measurements on:

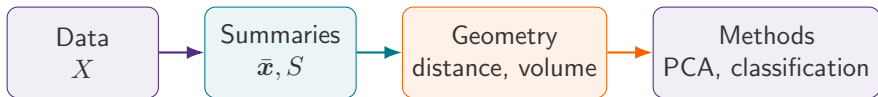
	X_1	X_2	X_3	X_4
Customer 1	88	90	76	80
Customer 2	95	94	89	85
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

We want to ask:

- ▶ Which customers are similar?
- ▶ Which variables contain similar information?
- ▶ How many dimensions are really present?

Nearly every answer will use matrix algebra

Roadmap for This Week



1. Build data with vectors and matrices
2. Summarize variation using covariance
3. Understand transformation, volume, direction, and distance
4. Connect the ideas to the methods used later in this course

One Observation Is a Vector

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix} \in \mathbb{R}^p$$

- ▶ $\mathbf{x}_i$ stores the p measurements for observation i
- ▶ x_{ij} is the value of variable j for observation i
- ▶ $\mathbf{x}_i$ is a $p \times 1$ column vector, and

$$\mathbf{x}_i^T = [x_{i1} \quad x_{i2} \quad \cdots \quad x_{ip}]$$

is a row vector

- ▶ Stacking all row vectors forms the data matrix X

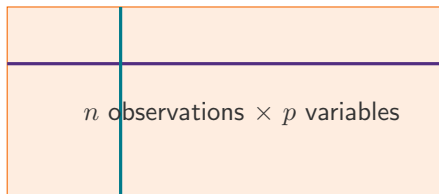
Where this appears later

Linear combinations, principal components, and canonical variates

Stack Observations to Form the Data Matrix

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}_{n \times p}$$

one row = $\mathbf{x}_i^T$



one variable

The Same Idea in R

```
X <- as.matrix(iris[, 1:4])  
# Create a 150 x 4 data  
  matrix  
dim(X)  
# Return the dimensions of X:  
  150 x 4  
X[1, ]  
# Extract the first  
  observation as a 1 x 4  
  row matrix  
X[, 1]  
# Extract the first variable  
  as a 150 x 1 column  
  matrix  
t(X)  
# Transpose X to obtain a 4 x  
  150 matrix  
crossprod(X)  
# Compute t(X) %*% X, a 4 x 4  
  cross-product matrix
```

Read dimensions before multiplying

If X is $n \times p$, then

$X^T X$ is $p \times p$, XX^T is $n \times n$

These matrices answer different questions

Quick check

Which product summarizes relationships among the *variables*?

Mean Vector and Covariance Matrix

Center of the Data

$$\bar{\mathbf{x}} = \frac{1}{n} \mathbf{X}^T \mathbf{1} = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{bmatrix}$$

One mean for each variable

Variation and Covariation

$$\mathbf{S} = \frac{1}{n-1} (\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}^T)^T (\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}^T)$$

One covariance for each variable pair

The Familiar Univariate Calculations

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}, \quad s_{jk} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

$\bar{\mathbf{x}}$ computes all p means, while $\mathbf{S}$ computes all $p \times p$ covariances simultaneously

How Should We Read a Covariance Matrix?

$$S = \begin{bmatrix} s_{11} & s_{12} & s_{13} \\ s_{21} & s_{22} & s_{23} \\ s_{31} & s_{32} & s_{33} \end{bmatrix}$$

- ▶ Diagonal: $s_{jj} = \text{Var}(X_j)$.
- ▶ Off-diagonal:
 $s_{jk} = \text{Cov}(X_j, X_k)$.
- ▶ Symmetric: $s_{jk} = s_{kj}$.
- ▶ Nonnegative definite:
 $\mathbf{a}^T S \mathbf{a} \geq 0$ (see next slide)

Where this appears later

PCA, Hotelling's T^2 , MANOVA, discriminant analysis

Covariance Is Variance After Projection

For any weights $\mathbf{a}$, define $Y = \mathbf{a}^T \mathbf{X}$. Then

$$\text{Var}(Y) = \text{Var}(\mathbf{a}^T \mathbf{X}) = \mathbf{a}^T \Sigma \mathbf{a} \geq 0$$

```
S <- cov(iris[, 1:4])
a <- c(0, 0, 1, 1)

# Two equivalent calculations
var(as.matrix(iris[, 1:4]) %*% a)
as.numeric(t(a) %*% S %*% a)
```

Interpretation

The quadratic form is not merely symbolic: it is the variance of a real linear combination of the variables

Identity and Inverse: Do and Undo

Identity: do nothing

$$I\mathbf{x} = \mathbf{x}$$
$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse: undo

$$\mathbf{x} \xrightarrow{A} \mathbf{y} \xrightarrow{A^{-1}} \mathbf{x}$$

$$A^{-1}A = AA^{-1} = I$$

No inverse exists when information collapses into a lower dimension

Where this appears later

Mahalanobis distance, multivariate normal likelihood, and discriminant analysis.

Inverse Matrices in R

Compute and check

```
S <- cov(iris[, 1:4])  
Sinv <- solve(S)  
  
round(S %*% Sinv, 6)  
kappa(S) #condition number
```

Interpretation

$$\text{solve}(S) = S^{-1}$$

$$SS^{-1} \approx I$$

A large condition number indicates that inversion may be numerically unstable

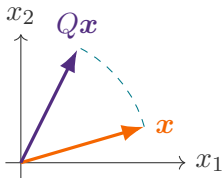
Quick check

If two variables are identical, can S be inverted?

Orthogonal Matrices Preserve Geometry

A square matrix Q is orthogonal when

$$Q^T Q = I \quad \implies \quad Q^{-1} = Q^T$$



$$\|Qx\| = \|x\|$$

$$(Qx)^T (Qy) = x^T y$$

Lengths and angles are preserved

**Think: rotation or reflection,
not stretching**

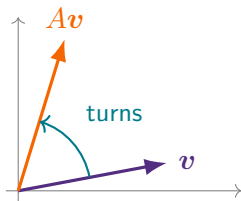
Where this appears later

PCA axes and factor rotation

Eigenvectors: Directions That Do Not Turn

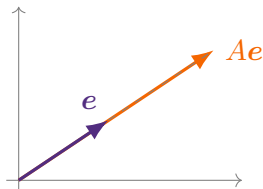
$$Ae = \lambda e, \quad e \neq 0$$

Ordinary direction



$$Av \neq \lambda v$$

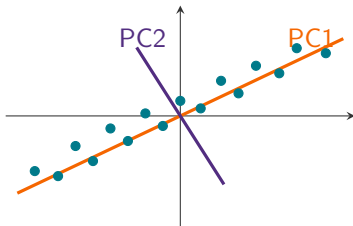
Eigenvector direction



$$Ae = \lambda e$$

Most directions turn; eigenvector directions are only rescaled

PCA Preview: Find the Dominant Directions



For covariance matrix S ,

$$S\mathbf{e}_j = \lambda_j\mathbf{e}_j.$$

- ▶ $\mathbf{e}_j$: principal direction
- ▶ λ_j : variance along that direction
- ▶ order $\lambda_1 \geq \dots \geq \lambda_p$

$$\text{PVE}_k = \frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^p \lambda_j}$$

Eigenvalues and PCA in R

```
# Center the four numerical variables
X <- scale(iris[, 1:4],
           center = TRUE, scale = FALSE)

# Sample covariance matrix
S <- cov(X)

# Eigen-decomposition of S
eig <- eigen(S)

eig$values           # variances along PCs
eig$vectors          # principal directions
eig$values / sum(eig$values) # proportion variance explained

# PCA directly in R
pca <- prcomp(X)
summary(pca)
plot(pca, type = "l")      # scree plot
```

$$Se_j = \lambda_j e_j$$

e_j : principal direction
 λ_j : variance along that direction

Spectral Decomposition: Rebuild the Matrix

For symmetric A ,

$$A = P\Lambda P^T = \sum_{j=1}^p \lambda_j \mathbf{e}_j \mathbf{e}_j^T.$$

$$P = [\mathbf{e}_1 \ \cdots \ \mathbf{e}_p], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p).$$

$$\mathbf{x} \xrightarrow{P^T} \text{eigenvector coordinates} \xrightarrow{\Lambda} \text{rescale} \xrightarrow{P} A\mathbf{x}$$

Rotate into eigen-directions $\rightarrow$ scale $\rightarrow$ rotate back

A symmetric matrix acts by scaling along orthogonal directions

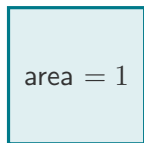
Trace, Determinant, and Rank Tell Different Stories

Quantity	Eigenvalue form	Multivariate meaning
Trace	$\text{tr}(A) = \sum_j \lambda_j$	Total marginal variation
Determinant	$ A = \prod_j \lambda_j$	Multidimensional volume; generalized variance
Rank	$\text{rank}(A) = \#\{\lambda_j \neq 0\}$	Effective linear dimensionality

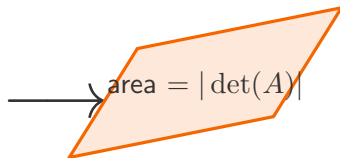
Quick check

Can two covariance matrices have the same trace but very different determinants?

Determinant Measures Area or Volume Scaling



unit square



transformed region

For a covariance matrix S , $|S|$ is called the **generalized variance**. It becomes zero when the data collapse into a lower-dimensional subspace

Correlation Shrinks Generalized Variance

For two variables,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} \implies |\Sigma| = \sigma_1^2\sigma_2^2(1 - \rho^2).$$

Weak correlation

The cloud occupies area in two directions; $|\Sigma|$ is relatively large.

Strong correlation

The cloud approaches a line; $|\Sigma|$ approaches zero.

Quick check

What happens when $|\rho| = 1$? What does this imply about Σ^{-1} ?

Trace, Determinant, and Rank in R

```
S <- cov(iris[, 1:4])
lambda <- eigen(S, symmetric = TRUE)$values

sum(diag(S))           # trace(S)
sum(lambda)            # same value

det(S)                 # generalized variance
prod(lambda)           # same value

qr(S)$rank             # numerical rank
```

Where this appears later

Scree plots, dimension selection, generalized variance, and singular covariance diagnostics.

Positive Definite Means Positive Variance

For symmetric A , the expression $x^T A x$ is a **quadratic form**

Nonnegative definite

$$x^T A x \geq 0 \quad \text{for all } x$$

All eigenvalues satisfy $\lambda_j \geq 0$.

Positive definite

$$x^T A x > 0 \quad \text{for } x \neq 0$$

All eigenvalues satisfy $\lambda_j > 0$.

Covariance matrices are always nonnegative definite; they are positive definite when no exact linear redundancy is present

Euclidean Distance Ignores Scale and Dependence

$$d_E^2(\mathbf{x}, \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^T (\mathbf{x} - \boldsymbol{\mu})$$

- ▶ One unit has the same meaning in every variable
- ▶ Correlation is ignored

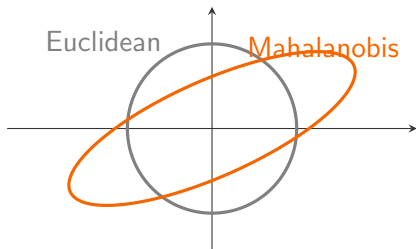
The problem

Is a 1-unit difference in income comparable to a 1-unit difference in age? Should movement along a highly correlated direction count as surprising?

Multivariate distance should adapt to the covariance structure.

Mahalanobis Distance Adapts to Covariance

$$d_M^2(\mathbf{x}, \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$



- ▶ Standardizes different scales.
- ▶ Accounts for correlation.
- ▶ Equal-distance contours are ellipses.

Where this appears later

Outlier detection, confidence ellipsoids, classification.

Mahalanobis distance in R

```
X <- as.matrix(iris[, 1:4])
center <- colMeans(X)
S <- cov(X)

d2 <- mahalanobis(X, center = center, cov = S)
head(d2)

# Flag observations beyond a reference cutoff
cutoff <- qchisq(.975, df = ncol(X))
which(d2 > cutoff)
```

Interpret with care

The chi-squared cutoff is motivated by multivariate normal theory; it is not a universal outlier rule

Ellipsoid Axes Come from Eigenvectors

If $A = P\Lambda P^T$, then

$$\mathbf{x}^T A \mathbf{x} = \sum_{j=1}^p \lambda_j (\mathbf{x}^T \mathbf{e}_j)^2 = c^2.$$

- ▶ Eigenvectors $\mathbf{e}_j$ determine the ellipsoid directions.
- ▶ Semi-axis lengths are proportional to $c/\sqrt{\lambda_j}$.
- ▶ Large λ_j means a shorter axis for the quadratic-form contour.

Eigenvalues connect algebra, geometry, distance, and probability.

Square-Root Matrices: Standardize and Reconstruct

For positive definite $A = P\Lambda P^T$,

$$A^{1/2} = P\Lambda^{1/2}P^T, \quad A^{-1/2} = P\Lambda^{-1/2}P^T.$$

Generate covariance

If z has covariance I , then

$$x = \mu + A^{1/2}z$$

has covariance A .

Remove covariance

If x has covariance A , then

$$z = A^{-1/2}(x - \mu)$$

has covariance I .

Square-Root Matrices in R

```
# Covariance matrix
S <- cov(iris[, 1:4])

# Spectral decomposition: S = P Lambda P'
eg <- eigen(S, symmetric = TRUE)
P <- eg$vectors
lambda <- eg$values

# Square root and inverse square root
S_half <- P %*% diag(sqrt(lambda)) %*% t(P)
S_inv_half <- P %*% diag(1 / sqrt(lambda)) %*% t(P)

# Check the key identities
round(S_half %*% S_half, 6)           # S
round(S_inv_half %*% S %*%
      S_inv_half, 6)                 # I
```

$$S^{1/2} = P\Lambda^{1/2}P^T, \quad S^{-1/2} = P\Lambda^{-1/2}P^T$$

$$S^{1/2}S^{1/2} = S \quad \text{and} \quad S^{-1/2}SS^{-1/2} = I$$

Partition a Random Vector

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}.$$

- ▶ Σ_{11} and Σ_{22} describe within-block variation.
- ▶ $\Sigma_{12} = \Sigma_{21}^T$ describes association between blocks.
- ▶ The dimensions must conform: if $\mathbf{X}_1$ is $q \times 1$, then Σ_{12} is $q \times (p - q)$.

Where this appears later

Conditional multivariate normal distributions and canonical correlation.

Concept Map: One Matrix, Many Methods

Building block	Where it appears later
Data matrix X	Every multivariate method
Mean vector $\bar{x}$	Multivariate normal modeling, group comparison
Covariance S	PCA, MANOVA, Hotelling's T^2
Inverse S^{-1}	Mahalanobis distance, discriminant analysis
Orthogonal matrix	PCA coordinates, factor rotation
Eigenpairs	PCA, factor analysis, canonical correlation
Trace / determinant	Total variance / generalized variance
Rank	Dimension reduction, singular covariance

Final Quick Check

Match each question to the most relevant concept

1. How much total marginal variation is present?
2. Which direction contains the most variation?
3. How far is a point after accounting for scale and correlation?
4. Has the data cloud collapsed to a lower-dimensional space?
5. How much multivariate volume does the covariance represent?

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1. trace
2. leading eigenvector
3. Mahalanobis distance
4. rank
5. determinant

Takeaways

- ▶ Rows of X are observations; columns are variables
- ▶ $\bar{x}$ and S summarize multivariate location and variation
- ▶ Eigenvectors provide directions; eigenvalues quantify variation along them
- ▶ Trace, determinant, and rank summarize total variation, volume, and dimension
- ▶ Quadratic forms turn covariance structure into meaningful distance